

Weekly Assignment: Sequences & Mathematical Induction (Epp 5.1 & 5.2)

Complete each of the following problems. Show all work and justify your answers.

Problem 1 (Epp 5.1 #2). Write the first four terms of the sequence defined by the formula:

$$b_j = \frac{5-j}{5+j}, \quad \text{for every integer } j \geq 1.$$

Problem 2 (Epp 5.1 #4). Write the first four terms of the sequence defined by the formula:

$$d_m = 1 + \left(\frac{1}{2}\right)^m, \quad \text{for every integer } m \geq 0.$$

Problem 3 (Epp 5.1 #44). Write the following expression using summation notation:

$$(1^3 - 1) - (2^3 - 1) + (3^3 - 1) - (4^3 - 1) + (5^3 - 1).$$

Problem 4 (Epp 5.2 #3). For each positive integer n , let $P(n)$ be the formula

$$1^2 + 2^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}.$$

- a. Write $P(1)$. Is $P(1)$ true?
- b. Write $P(k)$.
- c. Write $P(k+1)$.
- d. In a proof by mathematical induction that the formula holds for every integer $n \geq 1$, what must be shown in the inductive step?

Problem 5 (Epp 5.2 #4). For each integer n with $n \geq 2$, let $P(n)$ be the formula

$$\sum_{i=1}^{n-1} i(i+1) = \frac{n(n-1)(n+1)}{3}.$$

- a. Write $P(2)$. Is $P(2)$ true?
- b. Write $P(k)$.
- c. Write $P(k+1)$.
- d. In a proof by mathematical induction that the formula holds for every integer $n \geq 2$, what must be shown in the inductive step?

Problem 6 (Proof: Hoare Logic — proving $1 + 1 = 2$). Read the notes on Hoare logic in the course notes:

Recall the two key rules of Hoare logic:

- **Assignment axiom.** $\{Q[e/x]\} \ x := e \ \{Q\}$

To establish postcondition Q after assigning e to x , you need Q with every occurrence of x replaced by e to hold beforehand.

- **Sequencing rule.** If $\{P\} \ S_1 \ \{Q\}$ and $\{Q\} \ S_2 \ \{R\}$, then $\{P\} \ S_1; S_2 \ \{R\}$.

Prove the following Hoare triple:

$$\{x = 0\} \quad x := x + 1; \ x := x + 1 \quad \{x = 2\}$$

In other words, prove that $1 + 1 = 2$.

Hint: Work backwards from the postcondition. Apply the assignment axiom to the second assignment to find the intermediate assertion, then apply the assignment axiom to the first assignment and verify that it matches the precondition. Finally, combine both steps with the sequencing rule.